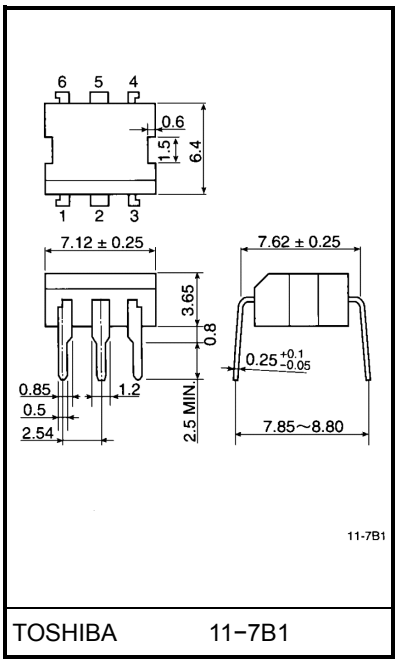
TOSHIBA Photocoupler GaAs Ired & Photo−Thyristor

TLP741G

Office Machine
Household Use Equipment
Solid State Relay
Switching Power Supply

Unit in mm



TOSHIBA 11−7B1

Weight: 0.35 g

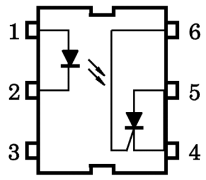
The TOSHIBA TLP741G consists of a photo-thyristor optically coupled to a gallium arsenide infrared emitting diode in a six lead plastic DIP package.

- Peak off−state voltage: 400V(min.)
- Trigger LED current: 10mA(max.)
- On−state current: 150mA(max.)
- UL recognized: UL1577, file no. E67349
- BSI approved: BS EN60065: 1994
  Certificate no. 6617
  BS EN60950: 1992
  Certificate no. 7366
- Isolation voltage: 4000V_{rms}(min.)
- Option (D4) type
  VDE approved: DIN VDE0884/08, 87
  Certificate no. 65640
  Maximum operating insulation voltage: 630V_{PK}
  Highest permissible over voltage: 6000V_{PK}

(Note) When a VDE0884 approved type is needed, please designate the "option (D4)"

	7.62mm pich standard type	10.16mm pich (LF2) type
• Creepage distance:	7.0mm(min.)	8.0mm(min.)
Clearance:	7.0mm(min.)	8.0mm(min.)
Insulation thickness:	0.5mm(min.)	0.5mm(min.)

Pin Configuration (top view)



1 : ANODE
2 : CATHODE
3 : NC
4 : CATHODE
5 : ANODE
6 : GATE

Maximum Ratings (Ta = 25°C)

Characteristic		Symbol	Rating	Unit
LED	Forward current	I_F	60	mA
	Forward current derating (Ta ≥ 39°C)	ΔI_F / °C	−0.7	mA / °C
	Peak forward current (100μs pulse, 100pps)	I_{FP}	1	A
	Power dissipation	P_D	100	mW
	Power dissipation derating (Ta ≥ 25°C)	ΔP_D / °C	−1.0	mW / °C
	Reverse voltage	V_R	5	V
	Junction temperature	T_j	125	°C
Detector	Peak forward voltage(R_{GK} = 27kΩ)	V_{DRM}	400	V
	Peak reverse voltage(R_{GK} = 27kΩ)	V_{RRM}	400	V
	On−state current	$I_{T(RMS)}$	150	mA
	On−state current derating (Ta ≥ 25°C)	ΔI_T / °C	−2.0	mA / °C
	Peak on−state current (100μs pulse, 120pps)	I_{TP}	3	A
	Peak one cycle surge current	I_{TSM}	2	A
	Peak reverse gate voltage	V_{GM}	5	V
	Power dissipation	P_D	150	mW
	Power dissipation derating (Ta ≥ 25°C)	ΔP_D / °C	−2.0	mW / °C
	Junction temperature	T_j	100	°C
Storage temperature range		T_{stg}	−55~125	°C
Operating temperature range		T_{opr}	−55~100	°C
Lead soldering temperature (10s)		T_{sol}	260	°C
Total package power dissipation		P_T	250	mW
Total package power dissipation derating (Ta ≥ 25°C)		ΔP_T / °C	−3.3	mW / °C
Isolation voltage (AC, 1 min., R.H. ≤ 60%)		BV_S	4000	V_{rms}

Recommended Operating Conditions

Characteristic	Symbol	Min.	Typ.	Max.	Unit
Supply voltage	V_{AC}	—	—	120	V_{ac}
Forward current	I_F	15	20	25	mA
Operating temperature	T_{opr}	−25	—	85	°C
Gate to cathode resistance	R_{GK}	—	27	33	kΩ
Gate to cathode capacity	C_{GK}	—	0.01	0.1	μF

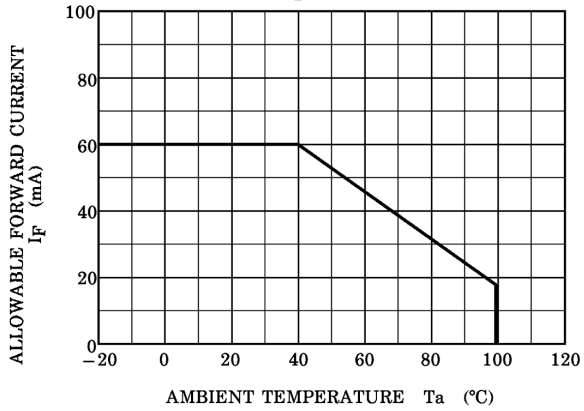
Individual Electrical Characteristics (Ta = 25°C)

Characteristic		Symbol	Test Condition		Min.	Typ.	Max.	Unit
LED	Forward voltage	V_F	I_F = 10mA		1.0	1.15	1.3	V
	Reverse current	I_R	V_R = 5V		—	—	10	μA
	Capacitance	C_T	V = 0, f = 1MHz		—	30	—	pF
Detector	Off−state current	I_{DRM}	V_{AK} = 400V, R_{GK} = 27kΩ	Ta = 25°C	—	10	5000	nA
				Ta = 100°C	—	1	100	μA
	Reverse carrent	I_{RRM}	V_{KA} = 400V, R_{GK} = 27kΩ	Ta = 25°C	—	10	5000	nA
				Ta = 100°C	—	1	100	μA
	On−state voltage	V_{TM}	I_{TM} = 100mA		—	0.9	1.3	V
	Holding current	I_H	R_{GK} = 27kΩ		—	0.2	—	mA
	Off−state dv / dt	dv/dt	V_D = 280V, R_{GK} = 27kΩ		5	10	—	V/μs
	Capacitance	C_j	V = 0, f = 1MHz	Anode to gate	—	20	—	pF
				Gate to cathode	—	350	—	

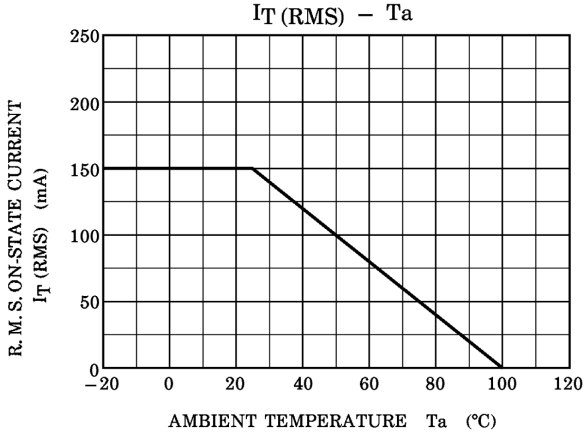
Coupled Characteristics (Ta = 25°C)

Characteristic	Symbol	Test Condition	Min.	Typ.	Max.	Unit
Trigger LED current	I_{FT}	V_{AK} = 6V, R_{GK} = 27kΩ	—	4	10	mA
Turn−on time	t_{ON}	I_F = 30mA, V_{AA} = 50V, R_{GK} = 27kΩ	—	10	—	μs
Coupled dv/dt	dv/dt	V_S = 500V, R_{GK} = 27kΩ	500	—	—	V/μs
Capacitance (input to output)	C_S	V_S = 0, f = 1MHz	—	0.8	—	pF
Isolation resistance	R_S	V_S = 500V	1×10^{12}	10^{14}	—	Ω
Isolation voltage	BV_S	AC, 1 minute	4000	—	—	V_{rms}
		AC, 1 second, in oil	—	10000	—	
		DC, 1 minute, in oil	—	10000	—	V_{dc}

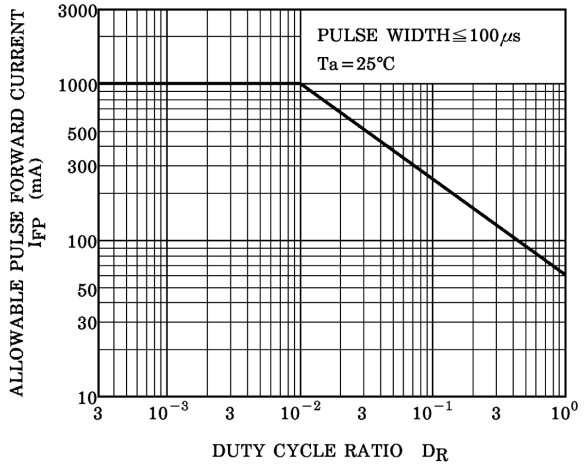
I_F – Ta



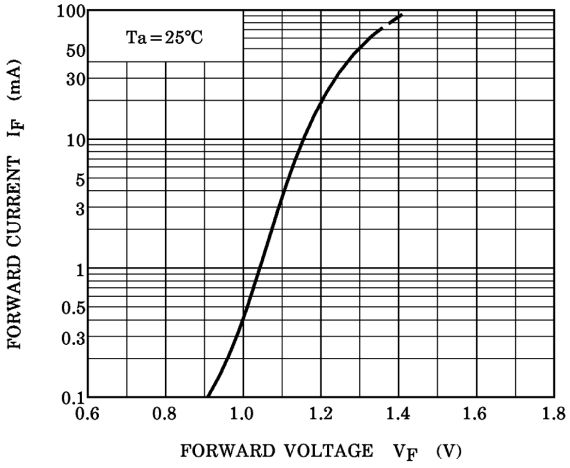
$I_{T\,(RMS)}$ – Ta



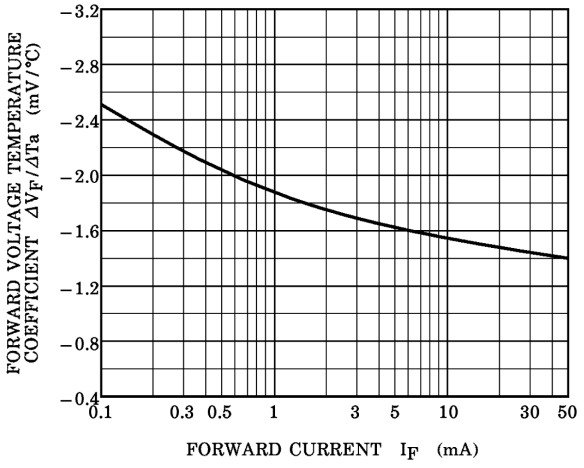
I_{FP} – D_R



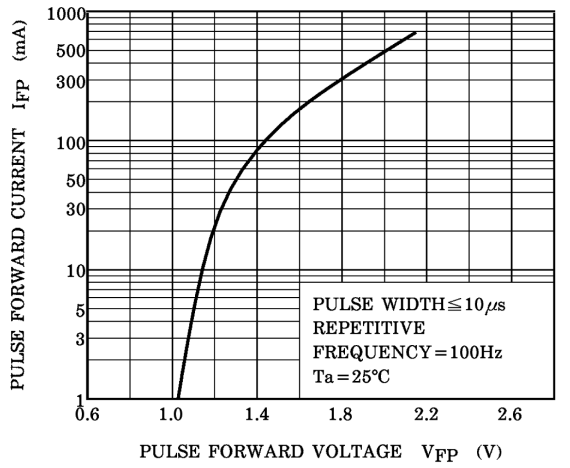
I_F – V_F



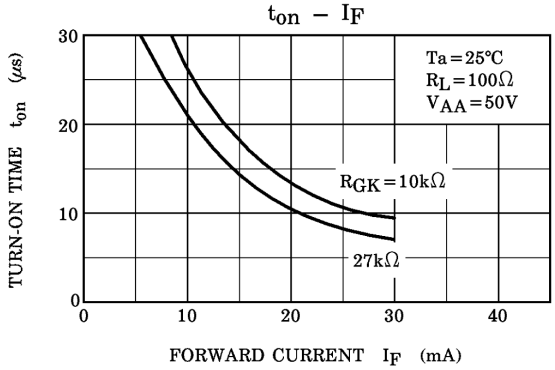
$\Delta V_F / \Delta Ta$ – I_F



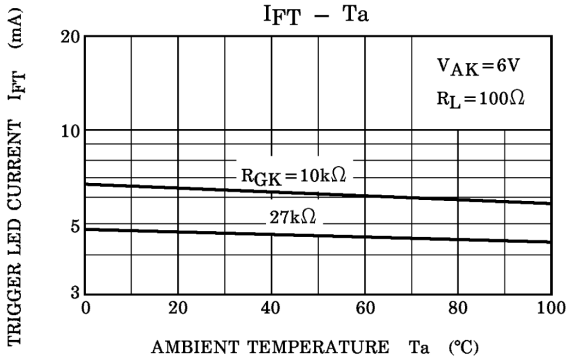
I_{FP} – V_{FP}



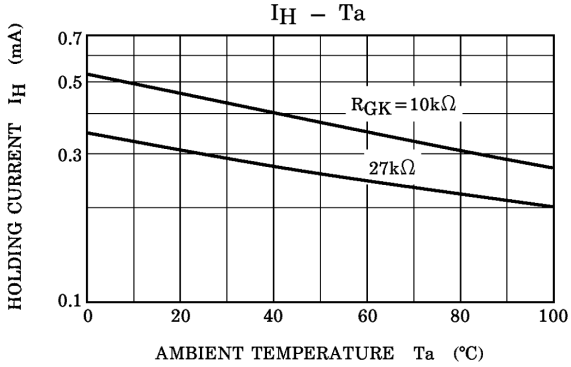
t_{on} – I_F



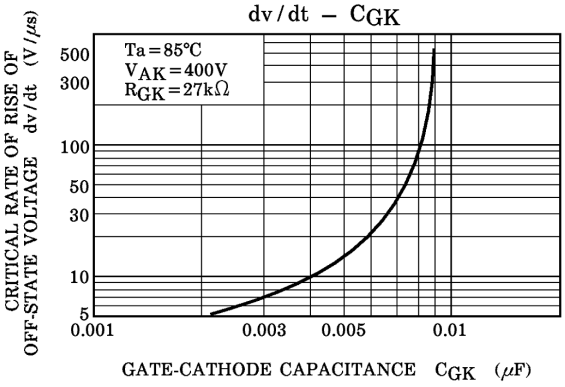
I_{FT} – Ta



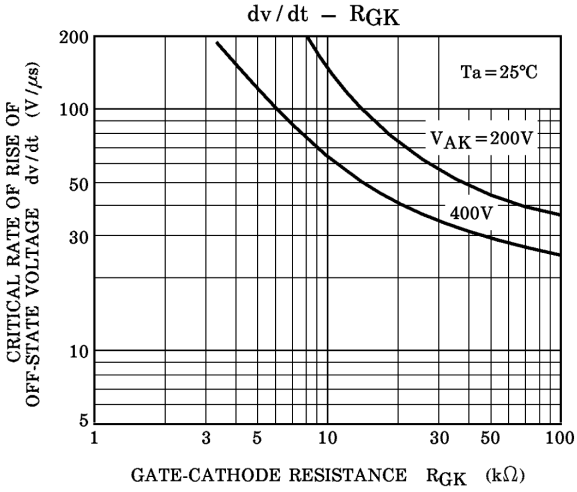
I_H – Ta



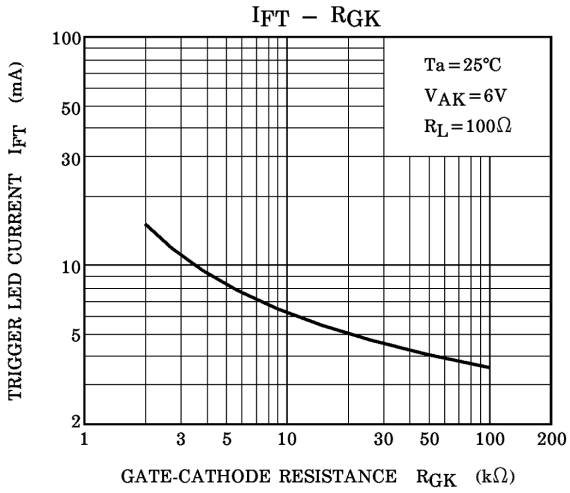
dv / dt – C_{GK}



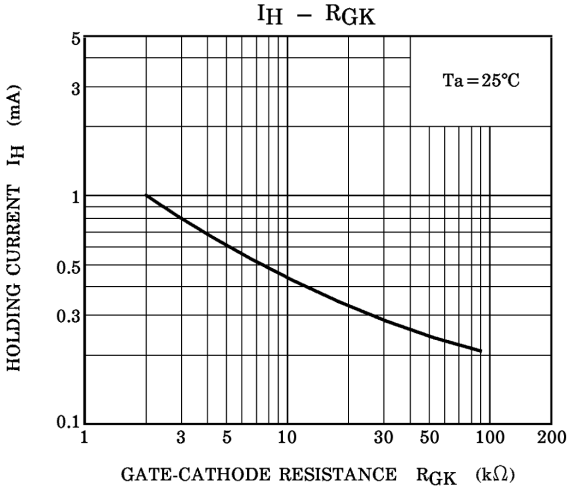
dv / dt – R_{GK}



I_{FT} – R_{GK}



I_H – R_{GK}



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